



infiniTy Online Camp JEE (Adv) 2024

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AREA UNDER CURVE

SUBJECT: MATHEMATICS

DATE: 05-04-2024

1. Find the area of the region $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$.
 (A) $\frac{19}{3}$ (B) $\frac{23}{3}$ (C) $\frac{19}{6}$ (D) $\frac{23}{6}$
2. If $f(x) = \begin{cases} \sqrt{\{x\}}, & x \notin \mathbb{Z} \\ 1, & x \in \mathbb{Z} \end{cases}$ and $g(x) = \{x\}^2$ (where $\{.\}$ denotes fractional part of x) then the area bounded by $f(x)$ and $g(x)$ for $x \in [0, 10]$ is
 (A) $\frac{5}{3}$ sq units (B) 5 sq units (C) $\frac{10}{3}$ sq units (D) None of these
3. Consider the functions $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$ and are defined as $f(x) = 2x - x^2$ and $g(x) = x^n$ where $n \in \mathbb{N}$. If the area between $f(x)$ and $g(x)$ is $\frac{1}{2}$, then n is a divisor of
 (A) 12 (B) 15 (C) 20 (D) 30
4. The area of the region between the curves $y = \sqrt{\frac{1 + \sin x}{\cos x}}$ and $y = \sqrt{\frac{1 - \sin x}{\cos x}}$ and bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$.
 (A) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$ (B) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$
 (C) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$ (D) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

5. Let $[x]$ denotes the greatest integer function. Draw a rough sketch of the portions of the curves $x^2 = 4[\sqrt{x}]y$ and $y^2 = 4[\sqrt{y}]x$ that lie within the square $\{(x, y) | 1 \leq x \leq 4, 1 \leq y \leq 4\}$. Find the area of the part of the square that is enclosed by the two curves and the line $x + y = 3$
- (A) $\frac{19}{3}$ (B) $\frac{23}{3}$ (C) $\frac{19}{6}$ (D) $\frac{23}{6}$
6. Let $f(x) = \begin{cases} -2, & -3 \leq x \leq 0 \\ x-2, & 0 < x \leq 3 \end{cases}$, where $g(x) = \min\{f(|x|) + |f(x)|, f(|x|) - |f(x)|\}$. Find the area bounded by the curve $g(x)$ and the X-axis between the ordinates at $x = 3$ and $x = -3$.
7. Let $f(x) = \min\left\{\sin^{-1} x, \cos^{-1} x, \frac{\pi}{6}\right\}, x \in [0, 1]$. If area bounded by $y = f(x)$ and X-axis, between the lines $x = 0$ and $x = 1$ is $\frac{a-X}{b(\sqrt{3}+1)}$. Then, $(a - b)$ is
8. Let f be a real valued function satisfying $f\left(\frac{x}{y}\right) = f(x) - f(y)$ and $\lim_{x \rightarrow 0} \frac{f(1+x)}{x} = 3$. The area bounded by the curve $y = f(x)$, the Y-axis and the line $y = 3$, is λe where $x, y \in R^+$ then $3\lambda =$
9. Area bounded by the curve $y = \max\{\sin x, \cos x\}$ and x-axis, between the lines $x = \frac{\pi}{4}$ and $x = 2\pi$ is equal to
- (A) $\frac{(4\sqrt{2}-1)}{\sqrt{2}}$ sq. units (B) $(4\sqrt{2}-1)$ sq. units
(C) $\frac{(4\sqrt{2}-1)}{2}$ sq. units (D) None of these
10. Area bounded by the parabola $y = x^2 - 2x + 3$ and tangents drawn to it from the point $P(1, 0)$ is equal to
- (A) $4\sqrt{2}$ sq. units (B) $\frac{4\sqrt{2}}{3}$ sq. units
(C) $\frac{8\sqrt{2}}{3}$ sq. units (D) $\frac{16}{3}\sqrt{2}$ sq. units
11. Area bounded by the curves $y = \sin x$, tangent drawn to it at $x = 0$ and the line $x = \frac{\pi}{2}$, is equal to
- (A) $\frac{\pi^2 - 4}{2}$ sq. units (B) $\frac{\pi^2 - 4}{4}$ sq. units
(C) $\frac{\pi^2 - 2}{4}$ sq. units (D) $\frac{\pi^2 - 2}{2}$ sq. units

12. If $A(n)$ represents the area bounded by the curve $y = n \cdot \ln x$, where $n \in N$ and $n > 1$, the x -axis and the lines $x = 1$ and $x = e$, then the value of $A(n) + nA(n - 1)$ is equal to
- (A) $\frac{n^2}{e+1}$ (B) $\frac{n^2}{e-1}$ (C) n^2 (D) en^2
13. Value of the parameter a such that the area bounded by $y = a^2 x^2 + ax + 1$, coordinate axes and the line $x = 1$, attains it's least value, is equal to
- (A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{4}$ (D) -1

ANSWER KEY												
Ques.	1	2	3	4	5	6	7	8	9	10	11	12
Ans.	D	C	BCD	B	C	11.5	3	9	A	C	B	C
Ques.	13											
Ans.	C											